

EFFECTS ON PHYSIOLOGICAL STRESS AND RECOVERY

 knowledgeiswatt.substack.com/p/106-double-threshold-days-effects

Knowledgeiswatt

October 8, 2025



Photo by [David Dvořáček](#) on [Unsplash](#)

In the last years, the **Norwegian Double-Threshold Days** have become very popular. This method consists of doing two daily sessions at medium/high intensity (the majority of Norwegians do them at an intensity between 2.0-4.0 mmol/L of blood lactate): one in the morning and one in the afternoon. As a consequence of the great success achieved by the Norwegian endurance athletes in the last years, performing multiple daily training session has become quite attractive also within the cycling world. So, it can be interesting to see some scientific data which can help us to learn more about this training strategy.

In the [last KIW Research Note](#) we saw that, when the total daily workload (both volume and intensity) is the same, multiple daily high intensity (HIT) training sessions seems to induce similar aerobic adaptations when compared to performing only one. However, splitting the same work into two shorter sessions could permit to accumulate less stress and fatigue. This could make possible to accumulate more medium/high intensity work within the microcycle.

When the total daily workload is the same, do double-threshold days induce different physiological stress and fatigue/time needed to recover when compared to condense the same work into one daily workout?

A study published by Kjøsén Talsnes and Colleagues (Centre for Elite Sports Research, Norwegian University of Science and Technology, Trondheim) in 2024 on *Frontiers in Physiology* tried to answer this question. (1)

WHAT DID THEY DO?

- 14 male endurance athletes (national-level cross-country skiers and runners) were recruited (age ~23 years, VO₂max ~70 ml/min/kg).
- They all completed the two following different training days in the lab matched for duration and external intensity:
 - **SINGLE SESSION DAY (SINGLE):**
 - 20min warm-up
 - 6x10min at 90% of speed at 4 mmol/L of blood lactate concentration (read high zone 3/low zone 4 in a 5/7 zones model) with 2min recovery in between
 - 15min cool down
 - **DOUBLE-THRESHOLD DAY (DOUBLE):**

The following workout was repeated two times within the same day (6.5 h of recovery in between):

 - 20min warm-up
 - 3x10min at 90% of speed at 4 mmol/L of blood lactate concentration (read high zone 3/low zone 4 in a 5/7 zones model) with 2min recovery in between
 - 15min cool down

To avoid any interference, participants were instructed to engage in low-intensity training exclusively and to replicate the same training regimen in the last two days before both SINGLE and DOUBLE.

The goal was to compare the physiological stress and time needed to recover induced by the two different training days.

WHAT DID THEY FIND?

-
- **Acute (during the sessions) physiological and perceptual stress was higher in SINGLE vs DOUBLE:**

This was due to a drift towards higher values that occurred in the second part of the SINGLE session. Here below the differences in physiological and perceptual stress markers in the second half of SINGLE when compared to the second workout of DOUBLE

- Heart Rate (bpm): $+4.2\% \pm 2.8\%$
 - Ventilation (L/min): $+4.9\% \pm 3.9\%$
 - Blood Lactate: $+0.91 \pm 0.88$ mmol/L
 - Rate of perceived exertion (RPE): $+1.0 \pm 0.7$ -point higher (range is 1-10)
- **In the morning of the subsequent day, levels of perceived fatigue and muscle soreness were higher following SINGLE compared to DOUBLE:**
 - perceived fatigue: $+1.0 \pm 1.5$ (scale was from 1 to 10)
 - perceived muscle soreness: $+1.0 \pm 1.5$ (scale was from 1 to 10)

PRACTICAL TAKEAWAYS:

When the total volume and intensity are matched, doing 2 medium/high intensity training sessions within the same day vs condensing all the work in one daily session:

- **induces less acute physiological stress and perceived effort and perceived training stress**, due to a drift in heart rate, blood lactate, ventilation and perceived effort in the second part of a longer single daily session.
- **generates lower levels of perceived fatigue and muscle soreness the following morning.**

GABRIELE'S COMMENT

These data are very interesting and, when added to what has been reported in the last KIW Research Note, they give some good suggestions about possible pros and cons of doing double instead of single daily medium/high intensity training sessions.

To give a brief summary, when total training duration and intensity (and so work done) are matched:

- aerobic adaptations are similar (VO₂max, muscle mitochondrial density)
- acute physiological stress (heart rate, blood lactate, ventilation, perceived effort) is lower when doing double sessions due to the drift that occurs in the second part of single longer daily session.
- double sessions days causes less fatigue the morning of the subsequent day.

How these information could be used?

- I think that overall, double medium/high intensity sessions per day can permits to accumulate more work at the same intensity and/or to work at a higher intensity given the lower 'physiological cost' associated with the two shorter sessions reported in this study. So, this method can be very interesting to improve performance in a rested state/in competitions with a relatively short duration (less than 2 hours?).
- On the other hand, condensing the work in only one longer daily session seems to stimulate more specifically durability (fatigue resistance, fatigue resilience) and so it could be more appropriate for athletes doing the specific work to prepare for longer competitions (>2 hours?).

These are just my speculations based on the last two KIW Research Notes.

Feel free to make also your own idea based on the data reported!

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

-
1. Kjøsén Talsnes R, Torvik PØ, Skovereng K, Sandbakk Ø. Comparison of acute physiological responses between one long and two short sessions of moderate-intensity training in endurance athletes. *Front Physiol.* 2024 Jul 30;15:1428536.





[14 Likes](#)